FreshPick model evaluation — 26 October 2026

The fine-tuned model performs well on the prepared AgriFreshNET test split, but these experiments do not support calling it reliable on general internet or camera images. It correctly predicted both produce and freshness on 752/777 dataset images (96.78%); it identified the correct produce in only 10/24 selected internet images (41.67%). Both original and fine-tuned checkpoints achieved the same web identity count. Do not describe the web result as measured freshness accuracy: no verified freshness labels exist for those images.

This audit uses FP32 offline batches of 64 with the deployed preprocessing on the RX 7800 XT. The earlier training report evaluated with AMP and obtained 96.65% accuracy / 0.9626 macro F1. FP32 obtains 96.78% / 0.9657; this is a numerical precision/batching difference with one additional correct prediction net, not another training improvement. The audit batch size differs from single-image serving; the full dataset numbers are batched offline measurements. All 24 web identity predictions were separately verified through the live single-image API. No weights, deployment rules, or thresholds were changed during this audit.

1. Scope and reproducibility

Compared the original and fine-tuned MobileNetV3-Small checkpoints on the same 777 test images (3,621 training / 776 validation). Evaluated 24 joint classes (eight produce types × three inherited freshness stages), collapsed produce identity, collapsed freshness, confidence calibration, four synthetic corruptions, and the running HTTP API. Same EXIF/RGB/aspect-preserving 224×224 black-padding/ImageNet normalization as the app. No retraining or test-driven model selection was performed.

Checkpoint: a9e3a06213d9c2442700a88bf3f970ebd3ded9941e7b3ecb19355e2e99a5cf24. Split digest: 97a36346183f44b03205bd884896f7fd072635727d67eaac3c45fed3e81e30d8. Fine-tuned model version: mobilenetv3-small-a9e3a06213d9c244.

Internet probes: three images per produce, 24 in total. Search candidates were visually reviewed and the selection manifest frozen before inference. Excluded dishes, illustrations, maps, the wrong species (e.g. tamarillo), and ambiguous mixed-produce scenes. Kept cultivar variation, multiple same-type produce, on-plant views, and four cut/open-fruit images as clearly disclosed challenges. Images and identity labels were checked against source metadata and visible content; there was one reviewer, no independent labeling panel, and no freshness ground truth. Internet-image overlap with ImageNet pretraining cannot be excluded. This is a convenience stress probe, not a representative unbiased benchmark.

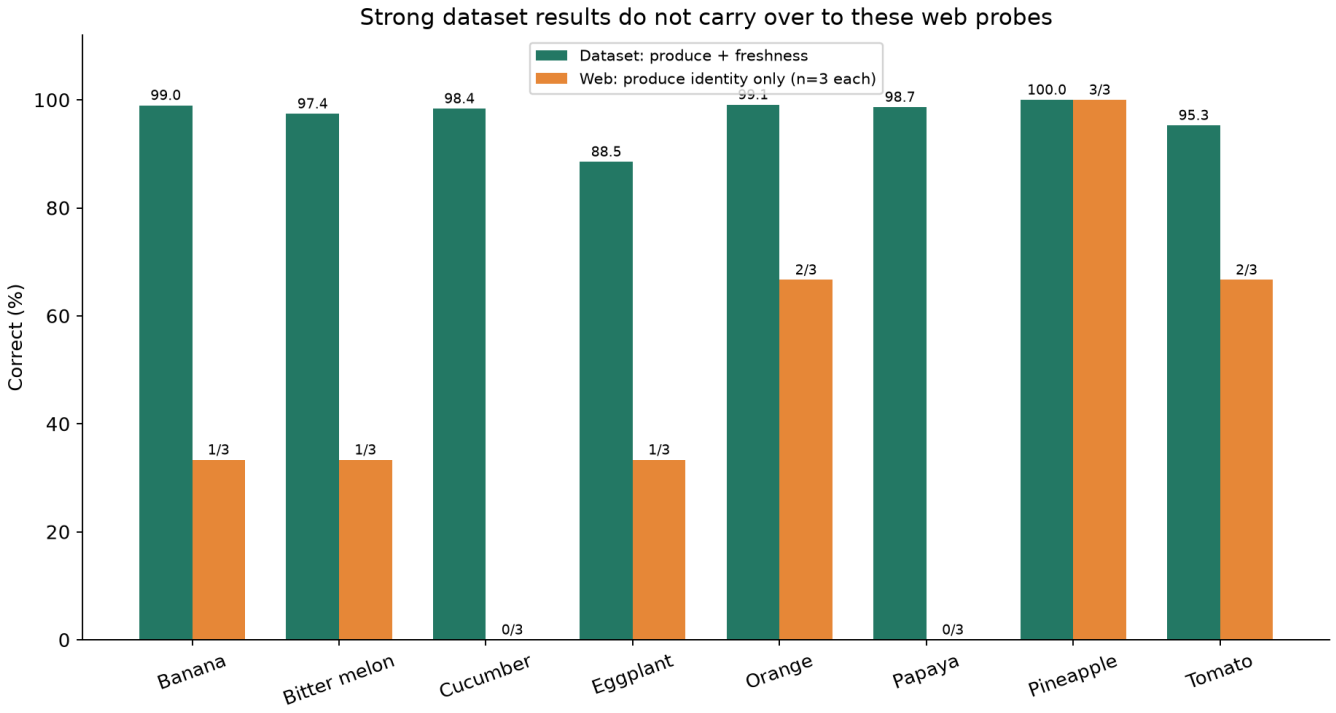
2. Overall measured results

Metric	Original	Fine-tuned
24-class accuracy	95.88%	96.78%
Macro precision	96.68%	97.25%
Macro recall / balanced accuracy	95.04%	96.18%
Macro F1	0.9572	0.9657
Weighted F1	0.9584	0.9676
Produce identity accuracy (dataset)	99.87%	100.00%
Freshness accuracy (dataset)	96.01%	96.78%
Joint NLL (lower better)	0.1206	0.0980
Joint Brier sum (lower better)	0.0649	0.0511
ECE (10 equal-width bins)	1.91%	1.49%
Web identity correct	10/24 (41.67%)	10/24 (41.67%)

3. Per-produce dataset performance and internet probes

Joint accuracy requires BOTH the right produce and freshness. Macro precision/recall/F1 average the three freshness-class metrics for that produce with equal weight; predictions into that produce from elsewhere count as false positives. All 777 produce identities were correct for the fine-tuned dataset pass, so remaining dataset errors are freshness confusions. Do not substitute the 100% produce identity result for freshness accuracy.

Produce	Dataset n	Joint correct	Joint accuracy	Macro precision	Macro recall	Macro F1	Web identity
Banana	100	99/100	99.00%	98.89%	99.26%	99.06%	1/3
Bitter melon	78	76/78	97.44%	98.04%	96.08%	96.91%	1/3
Cucumber	62	61/62	98.39%	99.05%	95.24%	96.95%	0/3
Eggplant	96	85/96	88.54%	89.84%	88.06%	88.46%	1/3
Orange	113	112/113	99.12%	99.10%	99.44%	99.26%	2/3
Papaya	75	74/75	98.67%	98.25%	95.24%	96.53%	0/3
Pineapple	83	83/83	100.00%	100.00%	100.00%	100.00%	3/3
Tomato	170	162/170	95.29%	94.84%	96.09%	95.39%	2/3



4. Uncertainty from small samples

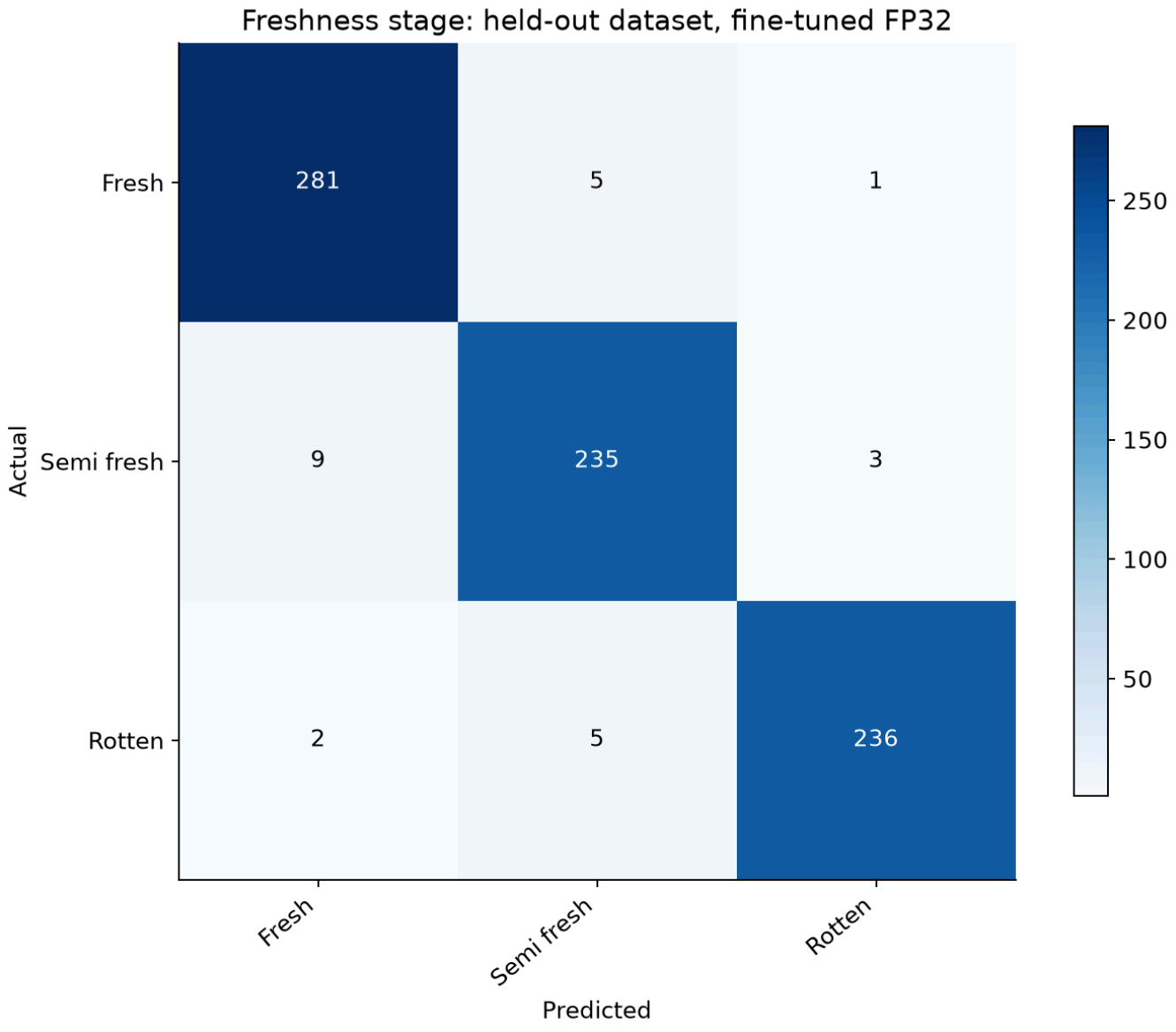
Intervals below are 95% Wilson intervals assuming independent trials. Related views violate that assumption, so even these intervals may be too optimistic. Three web images per produce cannot establish production accuracy. A 3/3 result has a lower bound of about 43.85%; 0/3 does not prove the model always fails.

Produce	Dataset accuracy	95% Wilson interval	Web identity	Web 95% Wilson interval
Banana	99.00%	94.55%–99.82%	1/3	6.15%–79.23%
Bitter melon	97.44%	91.12%–99.29%	1/3	6.15%–79.23%
Cucumber	98.39%	91.41%–99.71%	0/3	0.00%–56.15%
Eggplant	88.54%	80.64%–93.48%	1/3	6.15%–79.23%
Orange	99.12%	95.16%–99.84%	2/3	20.77%–93.85%
Papaya	98.67%	92.83%–99.76%	0/3	0.00%–56.15%
Pineapple	100.00%	95.58%–100.00%	3/3	43.85%–100.00%
Tomato	95.29%	90.99%–97.60%	2/3	20.77%–93.85%

5. Which freshness stages are missed?

Freshness is derived from the winning joint class, as in the UI. There are 2 inherited Rotten test labels predicted Fresh and 5 predicted Semi-fresh (243 Rotten examples total). These errors matter even when overall accuracy looks high. Eggplant is weakest overall: 85/96 joint predictions correct. Semi-fresh eggplant recall is 31/38 = 81.58%. Semi-fresh papaya and rotten cucumber each have only seven test examples; their class recall is unstable. In the offline audit, one test image has a winning-joint freshness stage different from the summed-marginal argmax; reported results follow the actual UI contract, not a substituted decision rule.

Stage	precision	recall	f1-score	support
fresh	96.23%	97.91%	97.06%	287
semi_fresh	95.92%	95.14%	95.53%	247
rotten	98.33%	97.12%	97.72%	243



6. All 24 joint classes

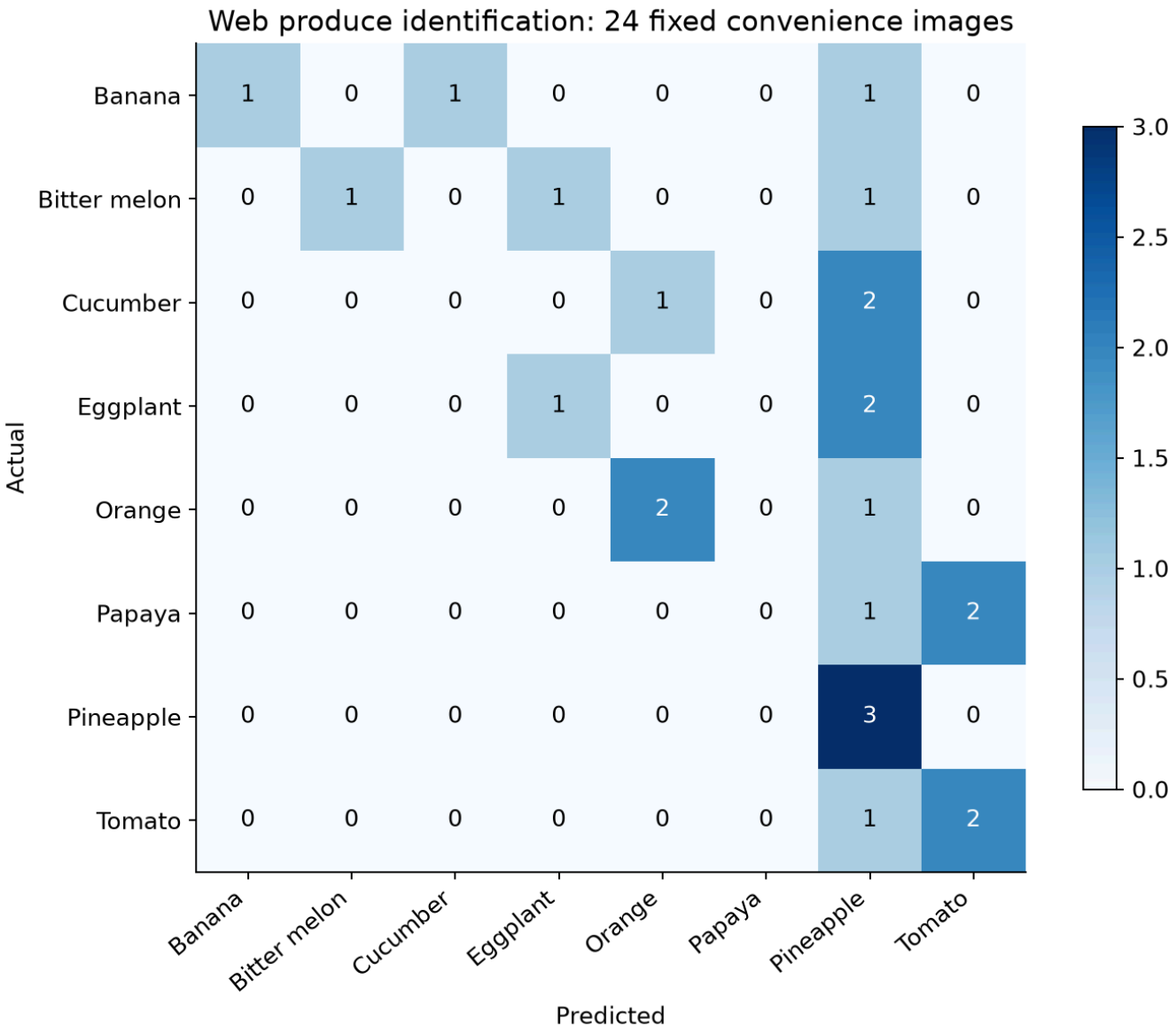
Class	Precision	Recall	F1	n
fresh_banana	100.00%	100.00%	100.00%	26
fresh_bittermelon	100.00%	100.00%	100.00%	29
fresh_cucumber	100.00%	100.00%	100.00%	21

Class	Precision	Recall	F1	n
fresh_eggplant	83.33%	100.00%	90.91%	35
fresh_orange	97.30%	100.00%	98.63%	36
fresh_papaya	94.74%	100.00%	97.30%	18
fresh_pineapple	100.00%	100.00%	100.00%	33
fresh_tomato	97.65%	93.26%	95.40%	89
rotten_banana	100.00%	97.78%	98.88%	45
rotten_bittermelon	94.12%	100.00%	96.97%	32
rotten_cucumber	100.00%	85.71%	92.31%	7
rotten_eggplant	95.00%	82.61%	88.37%	23
rotten_orange	100.00%	100.00%	100.00%	17
rotten_papaya	100.00%	100.00%	100.00%	50
rotten_pineapple	100.00%	100.00%	100.00%	32
rotten_tomato	97.30%	97.30%	97.30%	37
semi_fresh_banana	96.67%	100.00%	98.31%	29
semi_fresh_bittermelon	100.00%	88.24%	93.75%	17
semi_fresh_cucumber	97.14%	100.00%	98.55%	34
semi_fresh_eggplant	91.18%	81.58%	86.11%	38
semi_fresh_orange	100.00%	98.33%	99.16%	60
semi_fresh_papaya	100.00%	85.71%	92.31%	7
semi_fresh_pineapple	100.00%	100.00%	100.00%	18
semi_fresh_tomato	89.58%	97.73%	93.48%	44

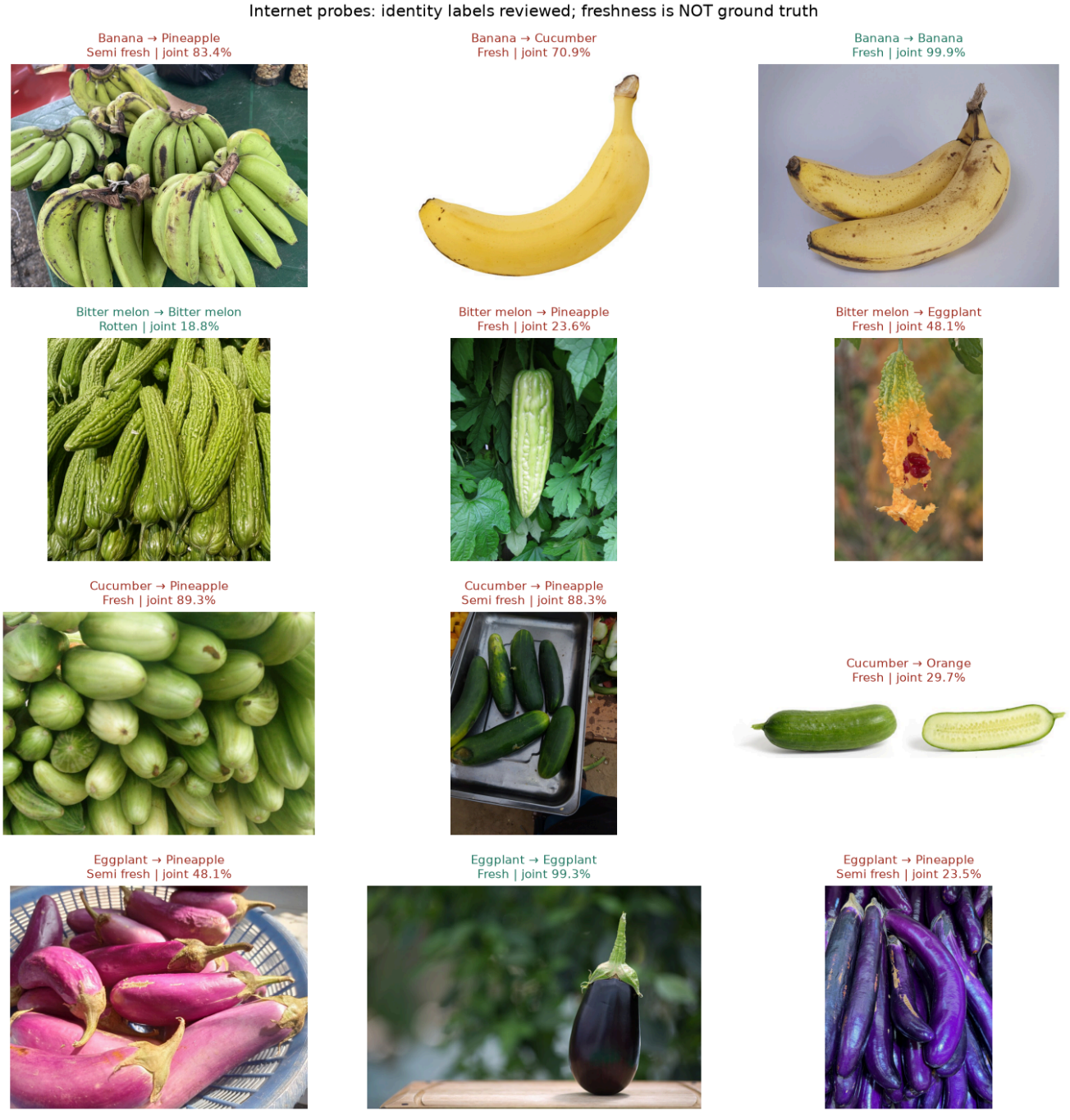
7. Internet experiment: identity only

Fine-tuned macro precision = 48.96%, macro recall = 41.67%, macro F1 = 0.3798. Pineapple was predicted for 12/24 images, but only three were actual pineapples: its web precision is 25%. Cucumber and papaya had 0/3 correct identities; pineapple had 3/3, which is too small to establish reliability. The internet freshness outputs are predictions to inspect, not correctness labels. Natural ripening, yellowing, dark spots, and open ripe fruit must not automatically be labeled Rotten.

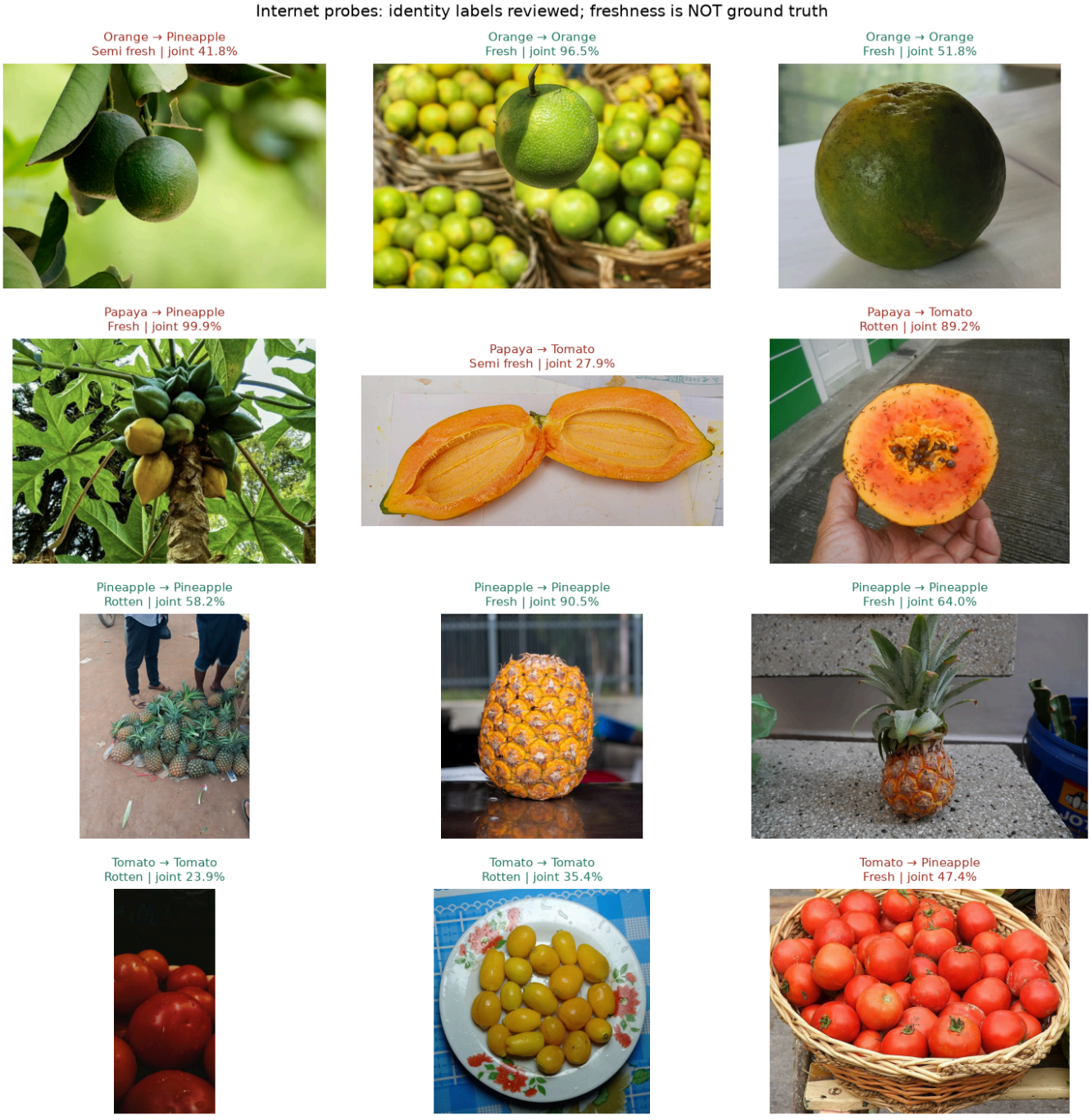
Produce	precision	recall	f1-score	support
banana	100.00%	33.33%	50.00%	3
bittermelon	100.00%	33.33%	50.00%	3
cucumber	0.00%	0.00%	0.00%	3
eggplant	50.00%	33.33%	40.00%	3
orange	66.67%	66.67%	66.67%	3
papaya	0.00%	0.00%	0.00%	3
pineapple	25.00%	100.00%	40.00%	3
tomato	50.00%	66.67%	57.14%	3


Web produce identification: 24 fixed convenience images
Actual
Predicted
Banana
Bitter melon
Cucumber
Eggplant
Orange
Papaya
Pineapple
Tomato
1 0 1 0 0 0 1 0
0 1 0 1 0 0 1 0
0 0 0 0 1 0 2 0
0 0 0 1 0 0 2 0
0 0 0 0 2 0 1 0
0 0 0 0 0 0 1 2
0 0 0 0 0 0 3 0
0 0 0 0 0 0 1 2
3.0
2.5
2.0
1.5
1.0
0.5
0.0

8. Inspect the internet predictions: first 12



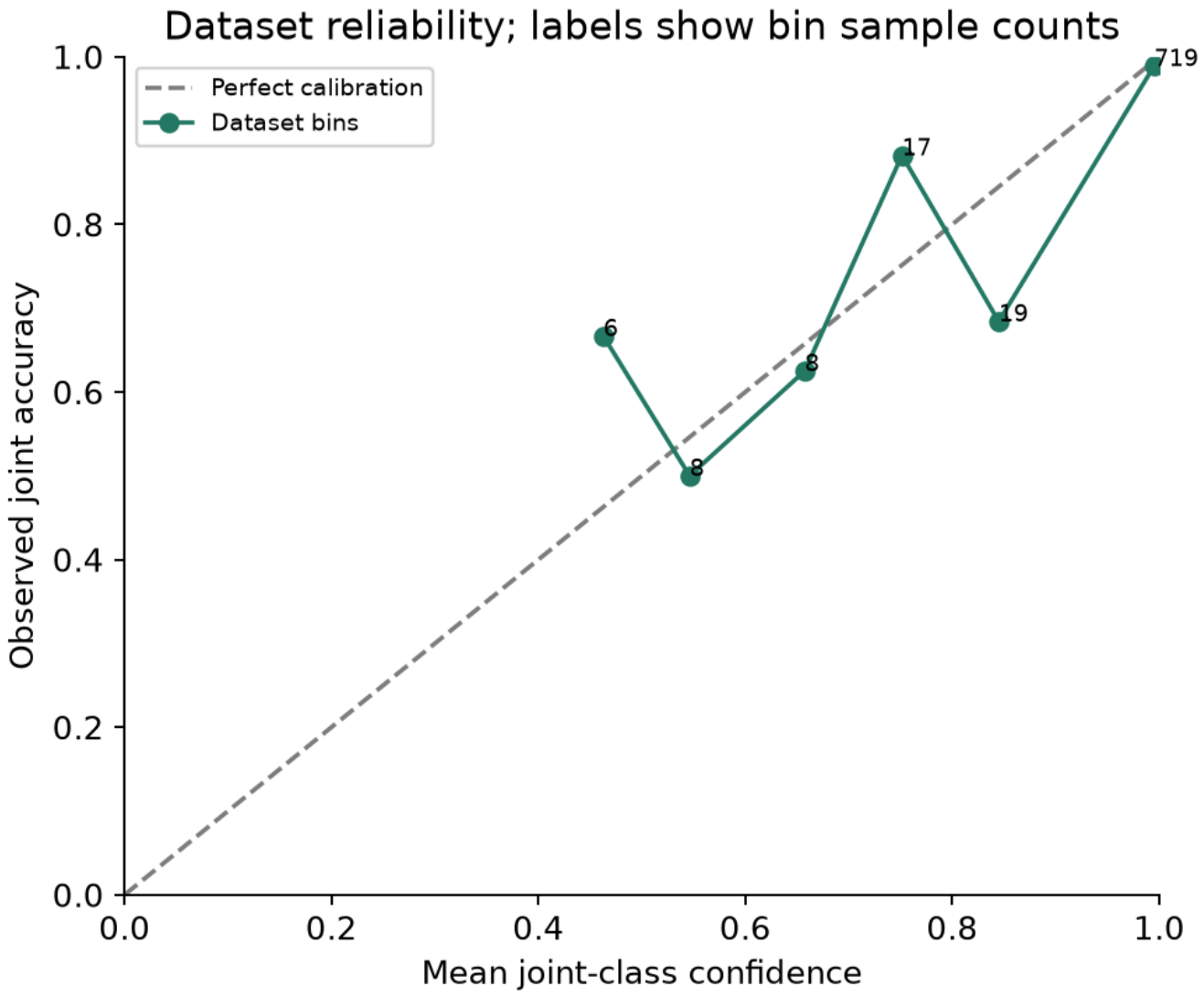
9. Inspect the internet predictions: remaining 12



10. Confidence does not transfer to the web

An on-tree papaya was classified as fresh pineapple with 99.92% winning-class confidence and 99.93% summed freshness confidence. Two cucumber photos were classified as pineapple with 89.34% and 88.32% joint confidence. A visually identifiable yellow banana on white background was classified as cucumber. These failures are consistent with domain/background/cultivar sensitivity; they do not prove which feature the network used. No validated feature explanation or causal background attribution was performed.

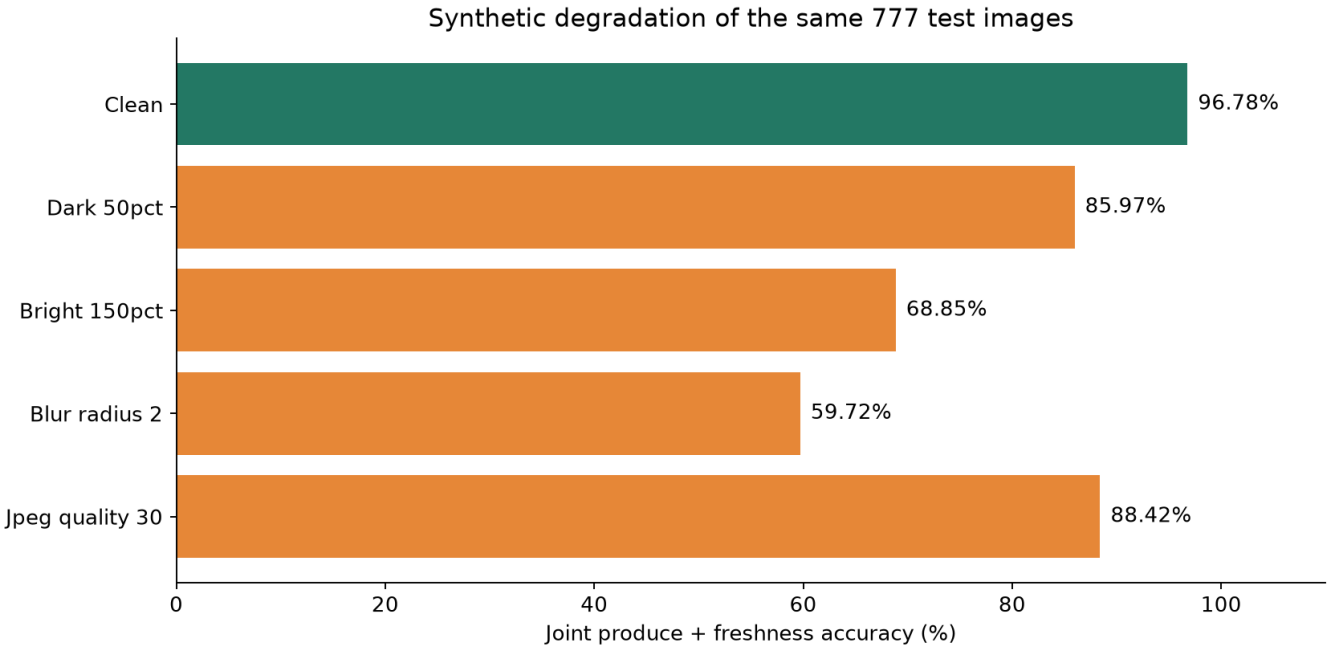
Dataset joint ECE is 1.49%, NLL 0.0980, and multiclass Brier sum 0.0511. These values only characterize this test distribution. Brier here sums squared errors over all 24 classes. Calibration bins are small outside the high-confidence group, and ECE alone can conceal subgroup failures. No temperature scaling or validated abstention threshold was fitted in this audit. A 90% confidence rule would still accept the papaya failure.



11. Synthetic image robustness

The same 777 prepared 224×224 images were degraded without changing labels: brightness multiplied by 0.5 or 1.5 (PIL clipping applies), Gaussian blur radius 2 pixels, and JPEG quality 30 before normal preprocessing. These are fixed stress settings, not estimates of the frequency or severity of real camera conditions. They demonstrate sensitivity and are not additional independent test samples.

variant	n	joint_accuracy	macro_f1
clean	777	96.78%	0.9657
dark_50pct	777	85.97%	0.8347
bright_150pct	777	68.85%	0.5962
blur_radius_2	777	59.72%	0.5431
jpeg_quality_30	777	88.42%	0.8623



12. Train/test similarity audit

No exact source-pixel hash overlaps between train and test were found. A 64-bit grayscale dHash nearest-neighbor screen found 33/777 test images within Hamming distance ≤3 of a training image and 117/777 within ≤5. Visual inspection of the eight closest pairs showed very similar backgrounds, poses, and produce appearance, consistent with related captures or near-duplicate views. Item identity cannot be proven without capture IDs, and dHash matches are not a measured leakage rate.

This reinforces the existing split limitation: random image splitting is not the same as testing on new produce items, sessions, cameras, or locations. The local evidence montage is runs/evaluation-web/nearest-train-test.jpg; pair paths and distances are in test-train-nearest-dhash.csv. Do not call a subset filtered by dHash a verified independent test.

13. Fine-tuning comparison is modest, not a proven domain improvement

On FP32 dataset inference the fine-tuned checkpoint gains seven correct predictions net: 19 original errors become correct while 12 previously correct cases become errors. The exploratory exact McNemar p-value is 0.281, which does not provide strong evidence of an improvement at 5% under independent paired-trial assumptions; related images and prior test reuse further limit that analysis. Web identity count stays at 10/24. More epochs on the same data are not the main next step.

14. Unsupported/nonproduce diagnostic

Four visually reviewed nonproduce images (shoes, an oil can, a street, and a clothing shop) were also sent to the running service. All four returned status ok and a supported produce label, as expected from a classifier with no unknown class. This is a diagnostic demonstration, not a measured general out-of-distribution rejection rate. The current service has no rejection gate.

id	predicted_produce	predicted_freshness	class_confidence	freshness_confidence	status
banana_more_01	cucumber	rotten	74.01%	76.15%	ok
banana_more_02	cucumber	rotten	41.59%	54.63%	ok
banana_more_03	papaya	rotten	78.00%	83.58%	ok
banana_more_05	pineapple	fresh	58.64%	74.57%	ok

15. Serving measurements

All 24 internet-image API predictions matched the offline fine-tuned produce predictions, and the server model version matched the checkpoint hash. HTTP upload/decode/preprocessing/GPU inference/JSON timing after three warmups: median 11.8 ms, p95 18.8 ms, range 6.9–21.6 ms over 24 sequential localhost requests. This excludes wide-area network latency, camera startup, concurrent load, and cold model loading. Speed is adequate for the prototype; data generalization is the pressing issue.

16. What to do next, in priority order

1. Build an independent camera benchmark before further tuning. Collect all eight produce types across devices, backgrounds, lighting, cultivars, and storage stages; retain item_id, capture_session, camera, site, and date. Assign all views/timepoints of one item to the same split. Keep a future collection/site/device untouched as a final test. Target at least 50 independent items per produce×freshness cell (1,200 item-level observations), then size the final study using the desired interval width; 50 is a pilot target, not a precision guarantee.
2. Improve labels. Write visible-condition criteria per produce and have two reviewers label images independently with disagreement adjudication. Distinguish unripe/ripe/overripe appearance from decay; mark ambiguous/occluded/cut/mixed-item images rather than forcing them into a freshness stage. Add a separate measurement protocol if shelf life is ever a goal.
3. Expand the training distribution. Add real phone photos, different backgrounds and cultivars, off-center objects, clutter, shadows, blur and compression. Prioritize eggplant freshness errors, cucumber/papaya identity failures, yellow/green cultivar variation, and the pineapple overprediction seen here. Add expert-reviewed negative/unsupported images. Keep this audit's web probes out of training until a new untouched benchmark is reserved.
4. Compare an object-crop/segmentation pipeline and a two-head produce+freshness model against the current 24-class baseline on identical grouped splits. A crop could reduce background sensitivity but must be measured, not assumed to solve it. Try brightness/color/blur/JPEG augmentation using training data only; keep realistic severity and verify per-produce recall.
5. Fit uncertainty behavior using validation data. Calibrate scores, evaluate selective accuracy versus coverage and per-produce coverage, and measure rejection on a held-out unsupported-image set. Low softmax confidence is not a verified unknown-item detector. Prevent a definitive assessment on rejected images; show a retake message. Do not deploy an arbitrary threshold based on these 24 probes.
6. Report the right release gates. Track joint macro F1, per-class recall with confidence intervals, especially Rotten recall and Rotten→Fresh mistakes; separately report produce identity, freshness, calibration, domain/slice robustness, and latency. Set acceptable errors and coverage with the intended users before training. Publish results from the untouched grouped benchmark, not just this reused random split.
7. Keep the current app clearly experimental until those checks pass. Show visible-condition limits, no shelf-life/food-safety claims, and no invented feature explanations. Do not turn confident failures into confident advice. A higher-scoring backbone is a later experiment after the data and evaluation protocol are fixed.

17. Metric definitions

Accuracy: fraction of predictions that match the label; joint accuracy requires both produce and freshness. Precision: TP/(TP+FP), answering how often a predicted class is correct. Recall: TP/(TP+FN), answering how many true examples were found. F1: harmonic mean of precision and recall. Support: number of true examples. Macro averages give each class equal weight; weighted averages weight by support. Multiclass balanced accuracy equals macro recall. Confusion matrices use actual classes as rows and predictions as columns. ECE compares confidence with correctness within bins; NLL penalizes assigning low probability to the true class. None of these is a food-safety probability. Web identity metrics do not measure freshness accuracy.

Source attribution and individual experiment records

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banana_direct_24025750	Assumulator	CC BY-SA 3.0
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bittermelon_more_04	Yun Huang Yong from Harbord, Australia	CC BY 2.0
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cucumber_direct_5765366	fir0002 flagstaffotos [at] gmail.com Canon 20D + Sigma 150mm f/2.8	GFDL 1.2
eggplant_01	GTSPACE	CC BY-SA 4.0
eggplant_04	Lufa Farms	CC BY-SA 2.0
eggplant_06	Oleg Yunakov	CC BY-SA 4.0
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papaya_01	Jovan Chris Barboza	CC BY-SA 4.0
papaya_04	Dinkun Chen	CC BY-SA 4.0
papaya_05	Judgefloro	CC0
pineapple_01	Grace789	CC BY-SA 4.0
pineapple_02	Wilfredor	CC0
pineapple_06	Zenyrgarden	CC BY-SA 4.0
tomato_01	Happie1Soul	CC BY-SA 4.0
tomato_03	SarKaLay တာလေး	CC BY-SA 4.0
tomato_04	KNOW MALTA by Peter Grima	CC BY-SA 2.0
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